

JEE MAIN REVISION MASTER: CIRCLE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 3 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Circle Equation, Tangents & Chord of Contact	3 PYQs	Circles
Standard	Tangents, Normals & Orthogonal Circles	0 PYQs	Tangents & Circles

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Circle Equation, Tangents & Chord of Contact	$r = \sqrt{g^2 + f^2 - c}$, $\text{Tangent: } y = mx \pm a\sqrt{1 + m^2}$, $\text{Chord of contact: } T = 0$, $\text{Orthogonal: } 2g_1g_2 + 2f_1f_2 = c_1 + c_2$
Tangents, Normals & Orthogonal Circles	$\text{Tangent: } y = mx \pm a\sqrt{1 + m^2}$, $\text{Length of tangent: } L = \sqrt{S_1}$, $\text{Orthogonal: } 2g_1g_2 + 2f_1f_2 = c_1 + c_2$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Circle Equation, Tangents & Chord of Contact (Moderate Yield — 3 PYQs)

Core Principle: Standard and general equations of circles, condition for line $y = mx + c$ to be tangent, and director circle $x^2 + y^2 = 2a^2$.

■ **Caution / JEE Pitfall:** The center of $x^2 + y^2 + 2gx + 2fy + c = 0$ is $(-g, -f)$, NOT $(+g, +f)$.

■ **Speed Trick:** The locus of the point of intersection of two perpendicular tangents to a circle is its director circle (radius $\sqrt{2}a$).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The area enclosed by the curve $x = \sqrt{3}\cos\theta, y = \sqrt{3}\sin\theta$ is..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "If in two circles, arcs of the same length subtend angles 30° and 78° at the centre, then the ratio of their radii is..." [Ans: N/A]
- [2018 Shift] Q#3 (MCQ): "The maximum area of a rectangle inscribed in the circle $(x+1)^2 + (y-3)^2 = 64$ is..." [Ans: N/A]

3.2 Tangents, Normals & Orthogonal Circles (Standard Yield — 0 PYQs)

Core Principle: Point and slope forms of circle tangents, director circle $x^2 + y^2 = 2a^2$, and orthogonality condition.

■ **Caution / JEE Pitfall:** The normal to a circle at any point ALWAYS passes through the center of the circle $(-g, -f)$.

■ **Speed Trick:** Common chord of two intersecting circles $S_1 = 0$ and $S_2 = 0$ is given directly by the linear equation $S_1 - S_2 = 0$.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "The area enclosed by the curve $x = \sqrt{3}\cos\theta, y = \sqrt{3}\sin\theta$ is..." [Ans: N/A]
- **[2024 Shift] Q#2 (MCQ):** "If in two circles, arcs of the same length subtend angles 30° and 78° at the centre, then the ratio of their radii is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The area enclosed by the curve $x = \sqrt{3}\cos\theta, y = \sqrt{3}\sin\theta$ is

- (A) $\sqrt{3}\pi$ sq. units
- (B) 9π sq. units
- (C) 6π sq. units
- (D) 3π sq. units

Correct Answer: None

Solution Analysis: Given, $x = \sqrt{3}\cos\theta, y = \sqrt{3}\sin\theta$ We know the standard parametric form of a circle is $x = a\cos\theta, y = a\sin\theta$. This represents a circle with centre at the origin and radius a . Here, $a = \sqrt{3}$. So the radius of the circle is $\sqrt{3}$. Area enclosed by a circle is πr^2 . So, $\text{Area} = \pi(\sqrt{3})^2$. $\text{Area} = 3\pi$. Hence, the area enclosed by the curve is 3π sq. units. Correct option: Option D.